Exp 0a – Lab Exploration

INTRODUCTION

In this experiment, you will learn where pieces of equipment are stored in the lab and get the opportunity to use some of the glassware you will be using for experiments throughout the semester. This will allow you to get comfortable with laboratory equipment, its purpose, and how to use it properly.

BACKGROUND

Buret

A buret is a specialty piece of glassware used for the purpose of delivering a volume of a liquid or solution to another vessel. Much like a graduated cylinder, the buret has graduations along the side used for measuring the volume of the liquid inside. A buret differs from a graduated cylinder in two important ways. First, there is a stopcock at the bottom of the instrument that can be opened to allow liquid to flow from the buret and closed again to stop the flow. The second important difference between a buret and a graduated cylinder is that the graduations are etched in decreasing order, instead of increasing order. Record the location of the bottom of the meniscus no matter how the graduations are ordered. Look at the example image below. The meniscus is between 46 mL and 47 mL. Begin at the smaller number (46.-- mL) and count the smaller lines (0.1 mL each) until the line is close to, but not past, the bottom of the meniscus (in this example, it is 46.8- mL). Estimate between the lines the location of the bottom of the meniscus to obtain an estimated digit (46.82 mL). The estimated digit is allowed to vary slightly from person to person. The volume should be read as 46.82 mL, NOT 47.18 mL. Holding a black bar (Figure 3) slightly below the bottom of the meniscus can help to highlight it and make it easier to read.

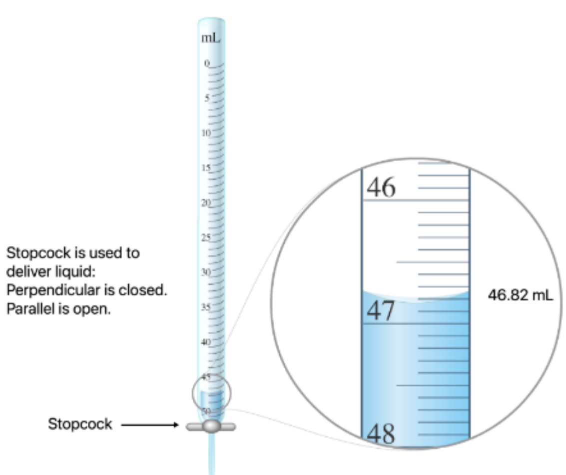


Figure 1:
Example of how to read a buret.

Volumetric Flask

Volumetric flasks are very accurate at measuring the volume of a liquid, but each volumetric flask can only measure one such volume. For example, the volumetric flask in the picture is a 10 mL volumetric flask. It can only contain 10 mL, but does so accurately to two decimal places, which must be recorded as 10.00 mL. This is the only volume that can be read from this particular flask. To use a volumetric flask, liquid is added until the bottom of the liquid's meniscus reaches the etching on the neck of the flask.

Volumetric flasks are most commonly used for preparing solutions. The solid (solute) is added to the flask first. Then the liquid (solvent) is added until the liquid level reaches the etching on the neck of the flask. It is recommended to first half-fill the flask, then swirl to dissolve your solid, before filling the rest of the way with solvent as it is much harder to mix once the flask is full.

Volumetric Pipet

A volumetric pipet is also a very accurate piece of glassware that is used to transfer a set volume from one vessel to another. Like a volumetric flask, it can only measure one volume, but does so very accurately, often to two decimal places. Liquid is drawn up through the pointed tip of the pipet, using a pipet bulb, until the bottom of the meniscus reaches the etching on the neck of the pipet. The pipet is then transferred to another vessel, and the liquid is allowed to drain out. Your instructor will show you how to properly work the pipet bulb, but slow and controlled is key. Going too quickly can cause you to accidentally draw solution into the bulb, which damages the bulb.

SUPPLY LIST

For this experiment, follow the procedure to gather items as needed.

- Buret
- Buret Clamp
- Bench Post
- Funnel
- Beaker (50-mL or 100 mL)
- Erlenmeyer Flask
- Scoopula/spatula
- Volumetric Flask (50-mL) × 2
- Volumetric Pipet
- Pipet bulb or pump
- Graduated Cylinder (100 mL)
- Wash bottle (filled with DI Water)

CLASSROOM SUPPLIES AND REACTANTS

These are shared supplies or reactants. Do not gather these items. Leave these items at their designated location in the classroom and use as needed according to the procedure.

- Food Coloring
- Sodium Chloride (Salt, NaCl)

PROCEDURE – PART 1: SCAVENGER HUNT

Use the boxes to check-off steps of the procedure as you complete each step.

1. ☐ Refer to the pictures on the "Equipment List" page of your lab manual (Page 3). This equipment is typically stored in the classroom on shelves or in drawers. Look around the room and collect the following items.
    a. Thermometer (in a white rack, usually located on the counter)
    b. Test tube
    c. Beaker (50-mL or 100-mL)
    d. Watch Glass
    e. Graduated Cylinder (100 mL)
    f. Erlenmeyer Flask
    g. Volumetric Flask (50-mL) + stopper
    h. Transfer Pipet
    i. Volumetric Pipet (10-mL)
    j. Pipet Bulb or Pipet Pump
    k. Buret
    l. Buret Clamp
    m. Bunsen Burner
    n. Funnel
    o. Wash Bottle
    p. Crucible Tongs
    q. Scoopula
    r. Ring Stands/Bench Posts (located in cabinets under fume hoods)
    s. Utility clamp (three finger clamp)
    t. Clamp holder (right angle rod clamp)

2. ☐ Arrange the items you found in the same arrangement they appear on the "Equipment List" page. Raise your hand to call over the instructor to have them sign off that you correctly found everything you needed. Alternatively, your instructor may request that you take a picture to submit with your data report.
3. ☐ Return all equipment that isn't listed in the "Supplies List" for this experiment and gather any other items listed in that section.

PROCEDURE – PART 2: USING A BURET

1. ☐ Assemble the titration apparatus as shown in the image to the right.
2. ☐ Add approximately 30 mL of DI water to a 50-mL beaker. Add one drop of food coloring. Swirl the beaker gently to mix the food coloring into the water.
3. ☐ Fill the buret with colored DI water from your beaker. You can use a funnel if you want to. The water level in the buret should be below the 0 mL mark at the top of the buret, and above the 10 mL mark. Never waste time trying to fill the buret to the 0 mL mark. If needed, drain some water or add additional water so that the water level is between the lines for 1-10 mL. Record the location of the bottom of the meniscus. ("Initial Buret Reading").
   a. Hold the black bar (Figure 3) behind the buret, slightly below the meniscus. This will help highlight the bottom of the meniscus to make it easier to read.
   b. Sketch the location of the meniscus on the diagram provided on the data report. Label ALL lines with the appropriate values.

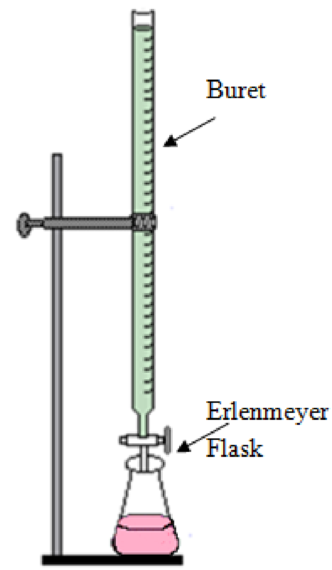


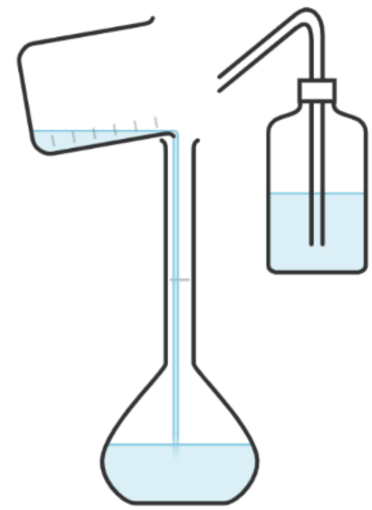
**Figure 2:
Titration Apparatus**

**Figure 3:
Hold this black bar behind the buret, below the meniscus to make it easier to read.**

4. ☐ Use a graduated cylinder to measure between 20-30 mL of DI water.
    a. Record the precise volume of DI water in the graduated cylinder on your report sheet.
    b. Sketch the location of the meniscus on the diagram provided on the data report (label the lines with the appropriate values).
    c. Transfer the measured DI water to an Erlenmeyer flask.
    d. Add a drop of food coloring to the flask. Use a different color than what is in your buret.
5. ☐ Place the Erlenmeyer flask under the tip of the buret. Open the stopcock of the buret so that it flows freely, while swirling the Erlenmeyer flask. Add approximately 5 mL of water from the buret, then close the stopcock.
6. ☐ Skills Checks: Watch your lab partner practice the following skills. When you have seen your lab partner successfully complete these skills, sign your name on their data report.
    a. Starting from a closed stopcock, very slowly start to open the stopcock until the water is flowing out of the buret dropwise. Add approximately 1 mL of liquid from the buret dropwise, then close the stopcock.
    b. Starting from a closed stopcock, very slowly start to open until only about half of a droplet has formed on the tip of the buret. Quickly close the stopcock. Use the wash bottle to rinse the droplet into your Erlenmeyer flask.
7. ☐ Record the final volume on the buret ("Final Buret Reading") and sketch the location of the meniscus on the diagram provided on the data report. Label the lines with the appropriate values.
8. ☐ Calculate the total volume of liquid delivered from the buret.

PROCEDURE – PART 3: USING PIPETS AND VOLUMETRIC FLASKS

1. ☐ Tare the balance by hitting the "0" or tare button so that it reads 0.0000 g.
2. ☐ Place an empty 50-mL **volumetric flask (with stopper)** on the balance.
3. ☐ Close the doors to the balance and try not to bump into the counter. Wait for the mass on the balance to stabilize. Record the precise mass on your report sheet (record all the digits that the balance display shows).
4. ☐ Remove the volumetric flask from the balance. Place a dry 50-mL **beaker** on the balance. Tare the balance by hitting the "0" or tare button so that it reads 0.0000 g again.



5. ☐ Use a scoopula to add between 4-6 g of sodium chloride (table salt, NaCl) to the beaker. Record the precise mass on the data report sheet.
6. ☐ Add approximately 10 mL of DI water to the beaker and swirl to wet the salt. Pour the wet salt into the 50-mL volumetric flask. Not all the salt will come out.
7. ☐ While holding the beaker's spout in the neck of the volumetric flask, spray a stream of DI water from a wash bottle into the beaker to rinse the remaining salt into the flask. Be careful not to spill or splash!
8. ☐ Rinse the beaker thoroughly into the flask to ensure all the salt is transferred, until the flask is about half to ¾ of the way full.
9. ☐ Place a stopper into the neck of the flask and swirl the volumetric flask until the sodium chloride crystals have dissolved.
10. ☐ Add one drop of food coloring.
11. ☐ Use a wash bottle to add more DI water to the volumetric flask until the bottom of the meniscus is aligned with the etching in the neck of the flask. You can use a plastic transfer pipet to carefully add water dropwise until exactly on the line.
12. ☐ Record the mass of the 50-mL volumetric flask now that it contains your prepared solution. This is your "Stock Solution."
13. ☐ Using a 10-mL volumetric pipet, transfer 10.00 mL of your stock solution into a new 50 mL volumetric flask. Make sure you do not draw liquid up into the pipet bulb or it will damage it.

14. ☐ Use a wash bottle to add more DI water to the volumetric flask until the bottom of the meniscus is aligned with the etching in the neck of the flask. You can use a plastic transfer pipet to carefully add water dropwise until exactly on the line. This is your "Diluted Solution."
15. ☐ Compare the color of your two solutions. Which one is darker? Which one is lighter?
16. ☐ Skills Check: When you have seen your lab partner successfully complete this skill, sign your name on their data report.
    a. Watch your lab partner use a 10-mL volumetric pipet to transfer 10.00 mL of solution to a new container.

CLEAN UP AND WASTE DISPOSAL

1. ☐ Pour all your salt solutions and water down the drain.
2. ☐ Wash your used glassware. There is soapy water near each sink.
3. ☐ Thoroughly rinse your glassware with DI water
4. ☐ You can blot the outside of the glassware dry with paper towels, but there is no need to thoroughly dry anything. You can put it back wet. Never attempt to dry the inside of narrow glassware with a paper towel!! Often, the paper towel gets stuck inside.
5. ☐ Return all the equipment.

Exp 0a (Intro) *Lab Exploration*

DATA AND REPORT – DUE ____ / ____ / ____

SECTION ____________________

DATE ______________________

PARTNER ___________________

PART 1: SCAVENGER HUNT

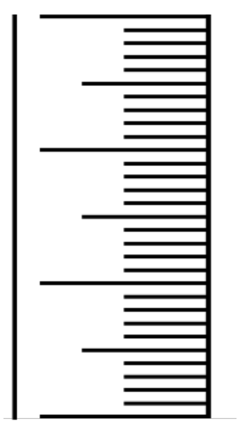
Instructor Check: __________________

DATA TABLE – PART 2: USING A BURET

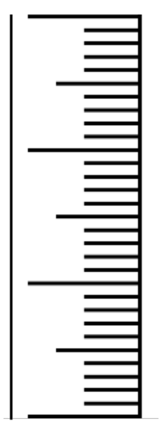
Measurements	
Graduated Cylinder	
Initial Buret Reading:	
Final Buret Reading:	
Calculations	
Volume Delivered by Buret	
Lab-Partner Check	
Adding Dropwise	
Adding Partial Drop	

SKETCH OF MENISCUS LOCATION

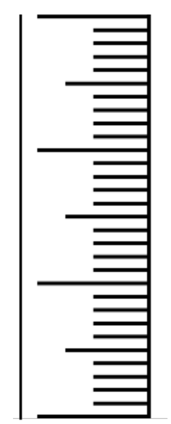
Graduated Cylinder



Initial Buret Reading:



Final Buret Reading:



DATA TABLE – PART 3: USING VOLUMETRIC FLASKS AND PIPETS

Measurements	
Mass of 50-mL Volumetric Flask + stopper (empty)	
Mass of sodium chloride (NaCl, salt)	
Mass of 50-mL Volumetric Flask + stopper + stock solution	
Visual comparison of stock solution and diluted solution	Observations:
Calculations	
Mass of stock solution made *(salt and water)*	
Mass of water used to make the stock solution	
Lab-Partner Check	
Pipet technique	